Arithmetic Sequences

Concept

An arithmetic sequence has a constant difference between consecutive terms.

General term

$a_n = a_1 + (n-1)d$; n is a positive integer.

Known terms

$d = (a_q - a_p)/(q - p)$.

Membership

A target is a term only if its index n is allowed.

Optional sum

$S_n = n(a_1 + a_n)/2$.

Example 1

Given $a_4 = 18$ and $a_{10} = 42$, find the general term.

Solution

1. $d = (42 - 18)/(10 - 4) = 4$.

2. $a_1 = 18 - 3(4) = 6$.

3. $a_n = 6 + 4(n - 1) = 4n + 2$.

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Example 2

Find term 40 of 12, 17, 22, ...

Solution

1. $a_1 = 12$ and $d = 5$.

2. $a_{40} = 12 + 39(5) = 207$.

Arithmetic Sequences

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Find d and a_{20} for $30, 27, 24, \ldots$.

2. If $a_3 = 11$ and $a_8 = 31$, find a_1.

3. Which term of $5, 12, 19, \ldots$ equals 103?

4. Optional: find the sum of the first 12 positive multiples of 3.

Arithmetic Sequences

Practice answers | check after trying

1. $d = -3$; $a_{20} = 30 + 19(-3) = -27$.

2. $d = (31 - 11)/5 = 4$; $a_1 = 11 - 8 = 3$.

3. $103 = 5 + 7(n - 1)$, so $98 = 7(n - 1)$, giving $n = 15$.

4. First 3, last 36, count 12: $S = 12(3 + 36)/2 = 234$.